

Jenkins Cheat Sheet

Quick revision for DevOps, Jenkins Pipeline and Jenkinsfile

1. Jenkins Basics

Concept	Meaning
Jenkins	Open-source automation server for CI/CD
Job / Project	A task configured in Jenkins
Build	One execution/run of a job
Node / Agent	Machine where Jenkins executes work
Controller	Coordinates Jenkins jobs and agents
Pipeline	CI/CD workflow defined as code
Jenkinsfile	Pipeline definition stored with source code
Stage	Logical phase such as Build, Test, Deploy
Step	Individual action such as sh, echo, junit

2. Common Jenkins Workflow

Step	Typical action
1	Developer pushes code to Git
2	Webhook / polling triggers Jenkins
3	Checkout source code
4	Build / compile
5	Run tests
6	Create artifact / Docker image
7	Deploy
8	Notify / archive results

3. Declarative Jenkinsfile — Must Know

Syntax	Example
Basic	<pre>pipeline { agent any stages { stage('Build') { steps { sh 'mvn package' } } stage('Test') { steps { sh 'mvn test' } } }</pre>
Agent	<pre>agent any agent none agent { label 'linux' }</pre>
Step	<pre>echo 'Hello' sh 'command' bat 'command'</pre>
Post	<pre>post { always { ... } success { ... } failure { ... } }</pre>
Environment	<pre>environment { APP_ENV = 'prod' }</pre>
Parameters	<pre>parameters { string(name: 'VERSION', defaultValue: '1.0') }</pre>
When	<pre>when { branch 'main' }</pre>

4. Frequently Used Pipeline Steps

Step	Use
echo	Print message

Step	Use
sh	Run Linux/macOS shell command
bat	Run Windows command
checkout scm	Checkout configured SCM
git	Clone/check out Git repository
junit	Publish JUnit test results
archiveArtifacts	Archive build artifacts
stash / unstash	Temporarily transfer files between stages
input	Manual approval
timeout	Limit execution time
retry	Retry a block
cleanWs	Clean workspace (plugin-dependent)

5. Environment & Build Variables

Variable	Meaning
BUILD_NUMBER	Current build number
BUILD_ID	Build identifier
BUILD_URL	URL of current build
JOB_NAME	Job name
WORKSPACE	Workspace path
BRANCH_NAME	Branch name in multibranch Pipeline
GIT_COMMIT	Current Git commit
currentBuild.result	Current build result
env.VARIABLE	Access environment variable
params.NAME	Access Pipeline parameter

6. Triggers & Cron

Trigger	Example
Poll SCM	pollSCM('H/5 * * * *')
Cron	cron('H 2 * * *')
Webhook	Common choice for GitHub/GitLab SCM events
Manual	Build Now / user-triggered build
Cron fields	MINUTE HOUR DOM MONTH DOW
Aliases	@hourly, @daily, @weekly, @monthly

7. Git + Jenkins Commands

Command	Purpose
git clone URL	Clone repository
git status	Check working tree
git add .	Stage changes

Command	Purpose
git commit -m 'msg'	Create commit
git push	Push changes
git pull	Fetch + integrate changes
git branch	List branches
git checkout branch	Switch branch
git merge branch	Merge branch

8. Docker + Jenkins

Pipeline	<pre>stage('Docker') { steps { sh 'docker build -t myapp:latest .' sh 'docker run -d -p 8080:8080 myapp:latest' } }</pre>
Key idea	Jenkins orchestrates; Docker packages/runs applications.
Credentials	Use Jenkins Credentials Manager; avoid hard-coding secrets.

9. Credentials & Security

Rule	Remember
Credentials	Store secrets in Jenkins Credentials Manager
Environment	Use credentials('credential-id') where appropriate
Never	Do not commit passwords, API keys or tokens into Jenkinsfile/Git
Least privilege	Give jobs only required permissions

10. Freestyle vs Pipeline

Freestyle	Configured mainly through Jenkins UI; simple jobs
Pipeline	Workflow represented as code; easier to version/review
Declarative	Structured, opinionated Pipeline syntax
Scripted	Groovy-based flexible Pipeline syntax

11. Interview / Exam One-Liners

CI	Continuous Integration — frequently integrate and test code
CD	Continuous Delivery/Deployment — automate software release flow
Jenkinsfile	Pipeline-as-code file, normally committed to SCM
Agent	Executes Pipeline work
Stage	Groups related steps
Step	Smallest executable Pipeline action
Webhook	External event can trigger Jenkins
Artifact	Build output stored/passed for later use
Multibranch Pipeline	Automatically discovers/builds branches and PRs
Blue Ocean	Jenkins UI experience focused on Pipeline visualization

Official reference: Jenkins Pipeline Syntax and Jenkins User Documentation. The Pipeline syntax supports Declarative and Scripted forms, and Jenkins recommends keeping a Jenkinsfile in source control.